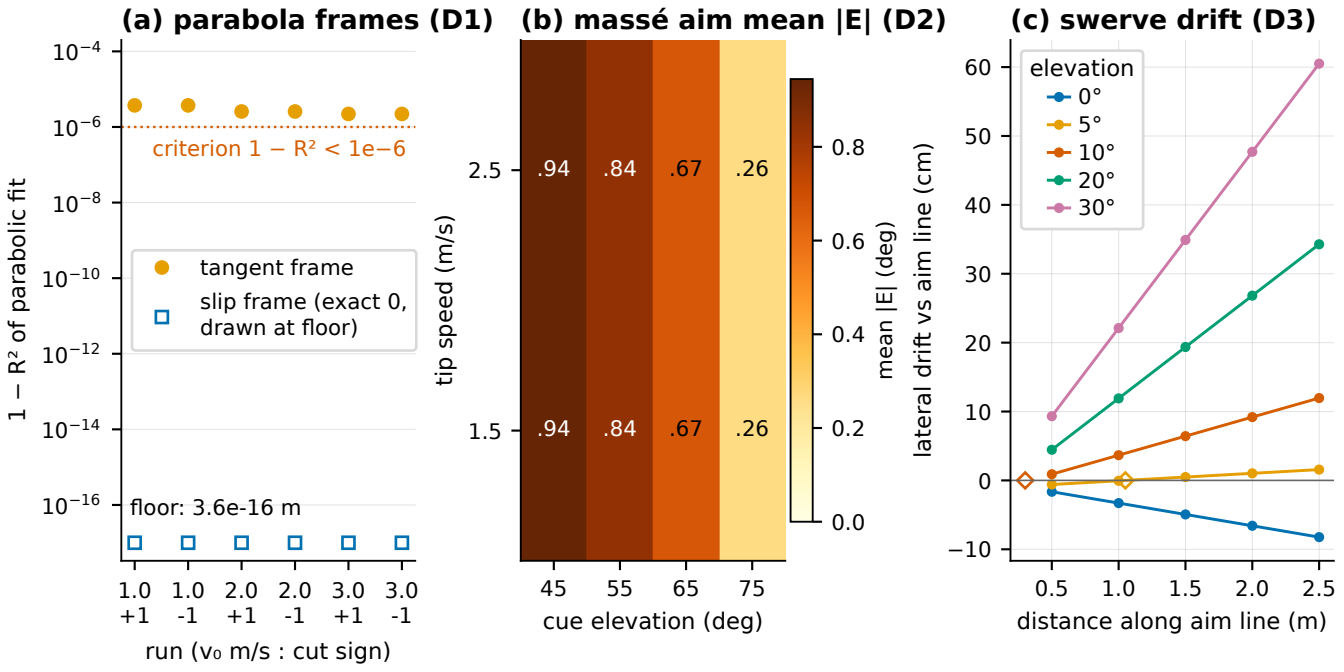
(a) parabola frames (D1)
(b) massé aim mean |E| (D2)
(c) swerve drift (D3)
1 − R² of parabolic fit
criterion 1 − R² < 1e−6
tangent frame
slip frame (exact 0, drawn at floor)
floor: 3.6e-16 m
1.0 +1
1.0 -1
2.0 +1
2.0 -1
3.0 +1
3.0 -1
run (v₀ m/s : cut sign)
tip speed (m/s)
2.5
1.5
.94
.84
.67
.26
mean |E| (deg)
cue elevation (deg)
elevation
0°
5°
10°
20°
30°
lateral drift vs aim line (cm)
distance along aim line (m)